Abie Thattamparambil

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GitHub | LinkedIn | Portfolio

EDUCATION

Toronto Metropolitan University Expected May 2027
Department of Electrical, Computer, and Biomedical Engineering - Biomedical Engineering | Toronto, ON

Relevant coursework: Signals and Systems I, II & III; Biomedical Instrumentation; Circuits, Devices and Sensors; Microprocessor Systems; Control Systems and Bio-Robotics; Electrical Machines and Actuators; Radiation Therapy Devices; Machine Learning and Analytics.

WORK EXPERIENCE

Reliability and Process Engineering Co-op Sep 2025 - Sep 2026
Sanofi Pasteur - Toronto Site | Toronto, Canada

- Completed **125+ FEAs** on obsolete spare parts in a cGMP-regulated vaccine manufacturing plant, validating material compatibility, equipment ratings, and NEMA/IP and ATEX electrical-safety requirements.
- Resolved Z4 discrepancies and corrected PMTLs/BOMs using OEM manuals and P&IDs; built Power BI/Excel dashboards to track NMI/Z4 items and prioritize work orders for shutdown execution and CAPAs.
- Co-developed a Poisson-distribution safety-stock model using four years of historical consumption data and multi-factor equipment criticality scoring.

Corporate Associate Jan 2025 - Present
Biomedical Engineering Course Union (BECU) | Toronto, Canada

- Coordinate industry outreach for lab tours, sponsorships, and career events; facilitate lab tours and information sessions connecting students with biomedical companies.

Marketing and Business Development Intern May - Aug 2022
MBG Corporate Services | Abu Dhabi, UAE

- Created an Excel tracking system for 100+ companies and used data visualization to monitor progress and inform management decisions.
- Supported targeted cold-call outreach, partnership discussions, technical presentations, and business-development negotiations.

PROJECTS

Blood Alcohol Concentration Breathalyzer using MQ-3 Sensor and Arduino Jan - Apr 2025

- Designed and built a low-cost breathalyzer using an MQ-3 alcohol sensor and Arduino to estimate BAC from BrAC readings and user input; calibrated sensor readings and implemented a voltage-divider circuit with a Wheatstone bridge to improve measurement accuracy.

Machine Learning Model for Kidney Cancer Detection Aug - Sep 2024

- Developed an image-classification model with **92% reported accuracy** using more than 10,000 CT and PET scans to distinguish tumour and normal cases; applied generative-AI techniques and data augmentation to increase dataset diversity and improve classification.

Motorized 3D-Printed Orthosis for Hand Rehabilitation Jan - Apr 2022

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